Venkata Lakshmi Parimala Pasupuleti

Backend / Distributed Systems · Python · AWS · Microservices · High Availability · Production at Scale · AI Systems

venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA · F-1 STEM OPT → H-1B

PROFESSIONAL SUMMARY

Backend engineer with 8+ years shipping production systems at scale - 100K+ monthly active users, 99.5% uptime, 20+ distributed service boundaries, and 1.6M-record data pipelines in production. At Entrata, led microservices architecture redesign (Docker, 50% scalability gain), defined integration architecture across 20+ service modules, and shipped Entrata's first production AI pipeline with compliance gates from day one. Built **AgentProbe** — a distributed 3-agent LangGraph backend (15-endpoint FastAPI, multi-provider routing, CI-integrated observability pipeline) running reliably at zero cost. Strong Python throughout; deep AWS (S3, EC2, RDS, Glue); hands-on with CI/CD, data pipeline architecture, and high-availability design. Excited to apply production reliability and systems thinking to AI infrastructure at a mission-driven team. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Distributed Systems | Backend Engineering | High Availability Design | System Orchestration | Reliability Engineering | Microservices Architecture | Python (Primary) | AWS Production | Data Pipeline Engineering | CI/CD & Observability | AI/ML Systems | Production at 100K+ MAU Scale

EXPERIENCE

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Distributed Systems · Microservices · AWS Data Pipelines · Production AI

- **Defined integration architecture across 20+ service boundaries** for a 100K+ MAU platform — led microservices redesign (Docker containerization achieving 50% scalability gain, 30% deployment time reduction); converted 20% of monolithic modules to service-based architecture with clean inter-service contracts.
- **Shipped 20+ production modules** (Python, PHP, PostgreSQL, RabbitMQ) across 120+ Agile sprints — sustained 99.5% uptime across 6+ consecutive months; led sprint planning, story estimation, and release management for a platform serving 100K+ monthly active users.
- **Built Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) — end-to-end ingestion, transformation, and output validation replacing manual Excel workflows; compliance gates and quality controls enforced at launch, zero violations.
- **Designed and operated data pipelines on AWS** (S3, EC2, RDS, Glue) — 100% PHI compliance migration using Libsodium-encrypted pipelines; zero security incidents; updated 50%+ of product codebase for encryption compliance across all data insertion and retrieval paths.
- **Optimized 200+ SQL queries** (30% performance gain) via execution plan tuning, indexing, and database partitioning; eliminated 80% of manual operational workload through Python/Bash automation; mentored 5+ engineers across 300+ peer code reviews.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Backend · Data Pipelines · AWS S3

- Built full-stack backend modules (PHP, PostgreSQL, JavaScript); designed SQL import scripts processing 1.6M HUD/Tax Credit records; migrated thousands of client documents to AWS S3 and redesigned legacy document storage database architecture.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer — Agentic Systems · Distributed Workflows · RAG

- Built a production agentic system as a **LangGraph stateful distributed workflow** (parse → plan → retrieve → validate → conditional routing) serving marketing, real estate, and commercial survey clients — compliance guardrails enforced deterministically at routing layer before any output fires.

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Quantitative Research · ML Anomaly Detection · Teaching (600+ students)

- Applied Isolation Forest to automate data integrity checks across 3 research pipelines; built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries.

PORTFOLIO PROJECTS

AgentProbe — Distributed Multi-Agent Backend with CI Observability Pipeline Distributed Systems Backend Observability

2024–Present · Python, LangGraph, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions · github.com/venkatalakshmiparimala/AgentProbe

- Built a **distributed 3-agent backend** (Orchestrator, RAG Tracer, Auto Fix) with inter-agent state handoffs, tool-use patterns, and multi-provider routing (Anthropic, OpenAI, Groq, Ollama) — 15-endpoint FastAPI backend, reliable at zero infrastructure cost.
- Designed a **CI-integrated observability pipeline**: 5-metric LLM-as-judge scoring on every commit via GitHub Actions — per-run telemetry, regression drift detection, retrieval quality, and latency tracked across concurrent agent test suites; regressions blocked before merge.
- Delivered **trace-level system observability** via React dashboard backed by ChromaDB RAG knowledge pipeline — real-time metrics across agent runs, storage, and retrieval layers.

Leasel — 8-Agent Distributed AI Platform with Multi-Layer Data Ingestion 8-Agent Pipeline System Orchestration Reliability

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a **multi-stage distributed pipeline**: OCR-fallback document ingestion → structured extraction → context-aware chunking → TF-IDF retrieval → 8-agent LangGraph StateGraph (real conditional edges) → deterministic validation gate → human escalation; each stage independently testable and replaceable.
- Enforced **reliability at every stage**: LLM reasons, deterministic logic decides — 100% citation correctness, 0 safety boundary violations on eval set; system validated end-to-end on misconfiguration root-cause scenario.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Backend & Distributed Systems: Python, Distributed Systems, Microservices, System Orchestration, High Availability, Reliability Engineering, Resilience, Scalability, Performance Engineering, Ingestion Pipelines, FastAPI, REST APIs, RabbitMQ, PostgreSQL, MySQL, Docker, CI/CD (GitHub Actions), Observability, Telemetry, Data Pipelines, ETL, Linux, Git

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Libsodium, Integration Architecture, Service-Based Architecture, Agile/Scrum

AI & ML Systems: LangGraph, LangChain, Multi-Agent Orchestration, PyTorch, Scikit-learn, Isolation Forest, RAG Pipelines, ChromaDB, LLM Evaluation, LLM-as-Judge, Prompt Engineering, OpenAI API, Anthropic API, NLP, Embeddings, Anomaly Detection, Feature Engineering

Languages: Python (primary), JavaScript, TypeScript, SQL, PL/SQL, PHP, R, Bash